

A JOURNEY THROUGH THE UNIVERSE

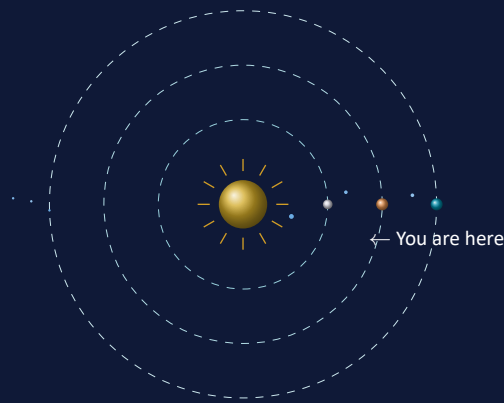
Astronomy

For Young Explorers

Ages 8–14

“Somewhere, something incredible is waiting to be known.”

— **Carl Sagan**



Discover planets and stars, rockets and galaxies,
the Moon and the Sun, and your place in the universe.

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Tawhid Bin Omar

Josephite Math Club • Version 1.0 • 23rd July 2026

A Journey Through the Universe

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PART I

Great discoveries begin with a single question: What is out there?

HAVE you ever looked up at the night sky and wondered what is out there? Those tiny twinkling lights. That big bright Moon. The dark space between them. Where does space begin? What lives there? And where do we fit in?

What is space, and where is Earth in the universe?

1.1 What Is Space?

Space begins about 100 kilometres above Earth. That is higher than any airplane flies! It is dark, cold, silent, and mostly empty. But space is also full of amazing things: stars, planets, moons, asteroids, comets, and galaxies.

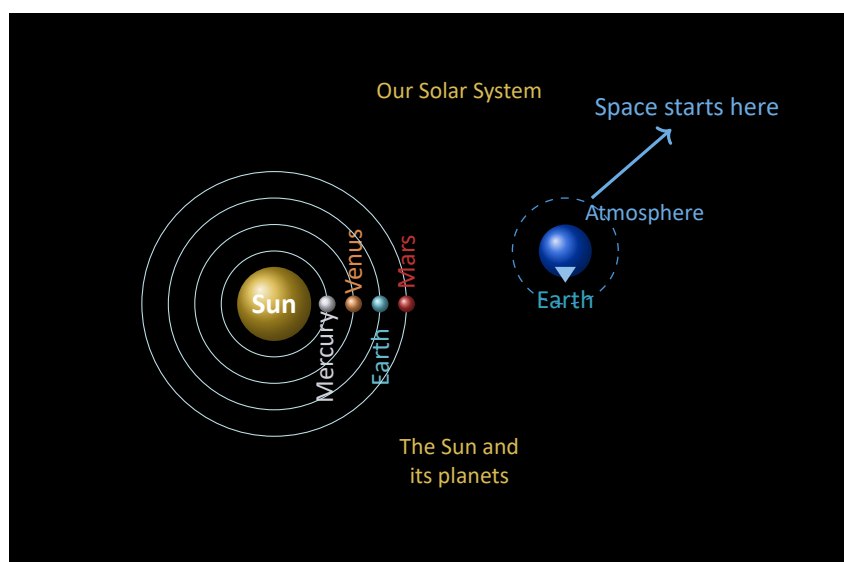


Figure 1.1: Earth is part of a solar system orbiting the Sun. Space begins just above our atmosphere.

Space

Space is the huge, mostly empty area beyond Earth's atmosphere. It contains everything: planets, stars, galaxies, and all the matter in the universe.

One Hour to Space

If you could drive a car straight up at 100 kilometres per hour, it would take only one hour to reach the edge of space!

1.2 We Live on a Planet

Earth is our home. It is a planet — a large round world that orbits the Sun. Earth is just right for life. It has water, air, and a temperature that is not too hot and not too cold.



Figure 1.2: Earth from space — the Blue Marble.

- Earth is the third planet from the Sun.
- It takes 365 days (one year) to travel around the Sun.
- About 71% of Earth's surface is covered with water.
- Earth is the only planet we know of that has life.

Invisible Shield

Earth has a magnetic field that protects us from the Sun's harmful particles. It acts like an invisible force field!

1.3 The Sun — Our Nearest Star

Every morning, the Sun rises and lights up the world. But what is the Sun? It is a star — a huge ball of hot, glowing gas. The Sun is the closest star to Earth, and it gives us light and warmth.



Figure 1.3: The Sun is enormous — more than a million Earths could fit inside it.

8-Minute Travel

Light from the Sun takes about 8 minutes to reach Earth. When you look at the Sun, you see it as it was 8 minutes ago!

The Sun Is a Star

All stars are giant balls of hot gas. Our Sun is a medium-sized yellow star. There are stars much bigger, much smaller, much hotter, and much cooler. At night, the stars you see are other suns — just very, very far away.

Never Look at the Sun

The Sun is so bright that looking at it can hurt your eyes forever. Never look directly at the Sun, even with sunglasses. Always use special solar filters or eclipse glasses.

1.4 Earth's Neighbour: The Moon

The Moon is Earth's closest neighbour in space. It orbits our planet once every 28 days. The Moon does not make its own light — it shines because sunlight bounces off its surface, like a mirror.

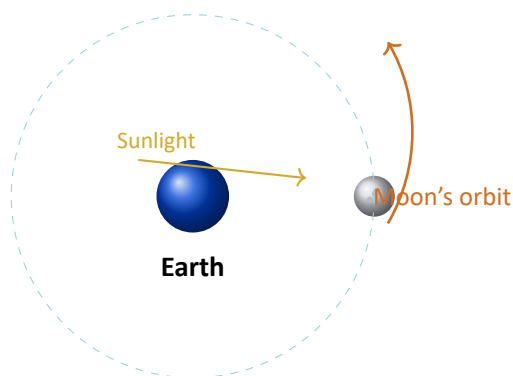


Figure 1.4: The Moon orbits Earth and reflects sunlight.

Always the Same Face

The Moon spins at exactly the same rate that it orbits Earth. That is why we always see the same side of the Moon!

Footprints Forever

The Moon has no wind or rain. The footprints left by Apollo astronauts in 1969 are still there today. They will last for millions of years!

1.5 Our Solar System Family

Our Sun has a family of eight planets orbiting around it. Together with their moons, dwarf planets, asteroids, and comets, they make up the solar system.

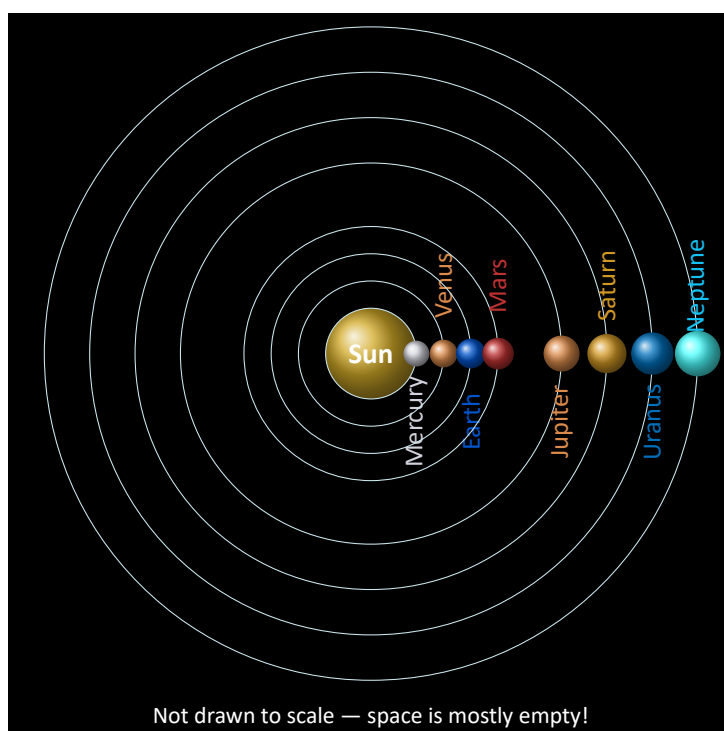


Figure 1.5: Eight planets orbit the Sun at different distances.

The Sun Is Everything

The Sun makes up 99.8% of all the mass in the solar system. All the planets, moons, and asteroids together make up only 0.2%!

1.6 The Night Sky

On a clear night, you can see hundreds of stars. They look tiny, but each one is a huge ball of hot gas, just like our Sun. They only look small because they are incredibly far away.

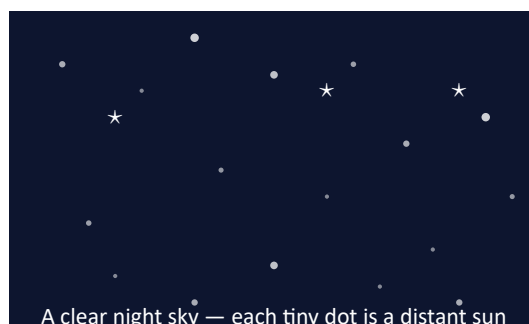


Figure 1.6: Every star you see is a sun, just like ours, far away.

Stars Twinkle

Stars appear to twinkle because their light passes through Earth's moving air. Actually, stars do not twinkle at all — it is our atmosphere that makes them shimmer!

Draw the Sky Tonight

Go outside after sunset with a grown-up. Look up and draw what you see. Include the Moon (what shape?), any stars, and anything else you notice. Write the date and time on your drawing. Try it again in a week. Has anything changed?

1.7 Chapter Review**Key Ideas**

- **Space** begins about 100 km above Earth. It is vast, mostly empty, and contains everything in the universe.
- **Earth** is a planet. It orbits the Sun and is the only known home for life.
- The **Sun** is a star — a giant ball of hot gas that gives us light and heat.
- The **Moon** orbits Earth. It shines by reflecting sunlight.
- Our **solar system** has the Sun, eight planets, and many smaller objects.
- **Stars** are distant suns. They look tiny because they are far away.

Quick Questions

1. Where does space begin?
2. What makes the Moon shine at night?
3. Why is the Sun important for life on Earth?
4. How many planets are in our solar system?
5. Why do stars look so small in the sky?
6. Can you name the planet we live on?

Words to Know

Atmosphere The layer of air around Earth.

Planet A large round world that orbits a star.

Star A huge ball of hot gas that makes its own light.

Orbit The path one object takes around another in space.

Solar System The Sun and everything that orbits it.

Universe Everything that exists — all space, matter, and energy.

Explore More

- **NASA Space Place:** <https://spaceplace.nasa.gov>
- **ESA Kids:** <https://www.esa.int/kids>

- **Watch the ISS:** <https://spotthestation.nasa.gov>
- **Stellarium (free planetarium):** <https://stellarium.org>



IMAGINE a huge family in space. At the centre is the Sun, a giant star. Orbiting around it are planets, dwarf planets, asteroids, and comets. Together, they make up our solar system — our cosmic home.

What objects live in our solar system, and how are they different?

2.1 The Eight Planets

There are eight planets in our solar system. In order from the Sun, they are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.

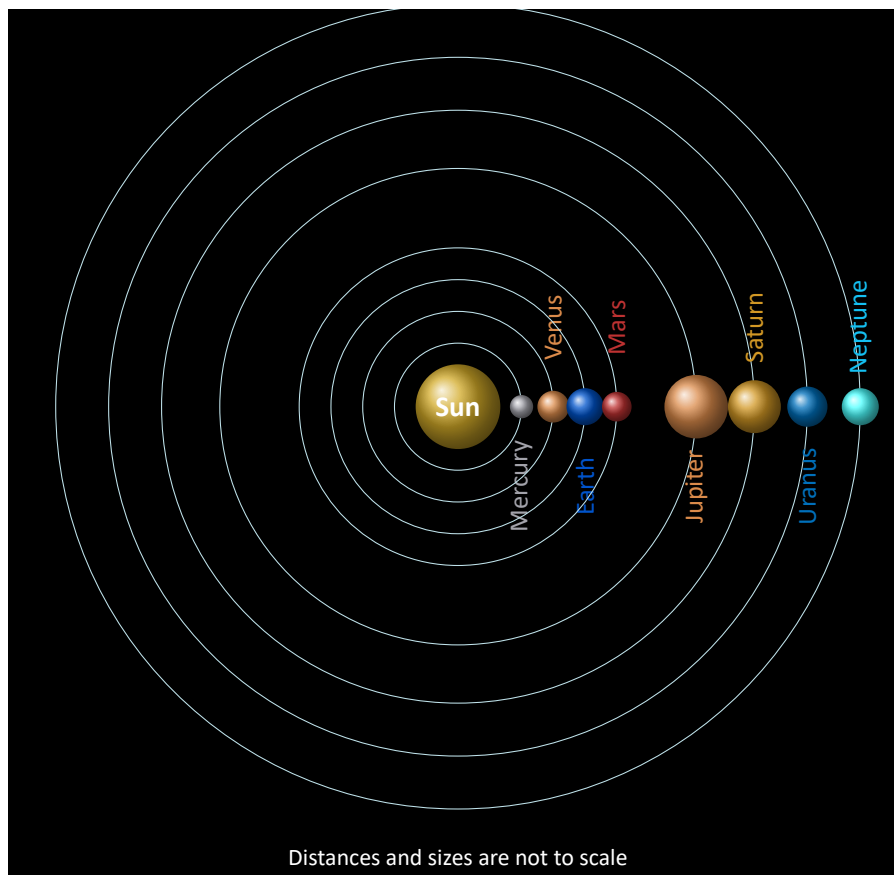


Figure 2.1: The eight planets orbit the Sun at different distances.

Planet

A planet is a large round world that orbits a star. Planets do not make their own light — they shine by reflecting their star's light.

Two Types of Planets

The planets come in two families:

Rocky Planets

Mercury, Venus, Earth, Mars

- Close to the Sun
- Small and rocky
- Solid surfaces
- Few or no moons

Gas Giants

Jupiter, Saturn, Uranus, Neptune

- Far from the Sun
- Large and gaseous
- No solid surface
- Many moons and rings

Planet Facts

Planet	Diameter	Day Length	Year Length	Fun Fact
Mercury	4,879 km	59 Earth days	88 days	Smallest planet
Venus	12,104 km	243 Earth days	225 days	Hottest planet
Earth	12,742 km	24 hours	365.25 days	Our home!
Mars	6,779 km	24.6 hours	687 days	Red planet
Jupiter	139,820 km	9.9 hours	11.9 years	Biggest planet
Saturn	116,460 km	10.7 hours	29.5 years	Has amazing rings
Uranus	50,724 km	17.2 hours	84 years	Spins on its side
Neptune	49,244 km	16.1 hours	165 years	Fastest winds

Mercury to Neptune

A fun way to remember the planets: **My Very Eager Mother Just Served Us Noodles!**

2.2 A Closer Look: The Inner Planets**Mercury**

The smallest planet and closest to the Sun. It is covered in craters and has almost no atmosphere. Days are scorching hot (430°) and nights are freezing (−180°).

Venus

The hottest planet! Thick clouds of acid trap heat, making its surface 465°. Venus spins backwards compared to most planets. A day on Venus is longer than its year!

● Earth

Our home! The only planet known to have life. Liquid water covers 71% of its surface. Earth's perfect distance from the Sun makes it a Goldilocks planet — not too hot, not too cold.

● Mars

The red planet. It has the largest volcano in the solar system — Olympus Mons, three times taller than Mount Everest. Mars has ice caps at its poles, just like Earth!

2.3 A Closer Look: The Outer Planets

● Jupiter

The biggest planet. Its Great Red Spot is a storm bigger than Earth that has been raging for hundreds of years. Jupiter has 95 known moons!

● Saturn

Famous for its beautiful rings made of ice and rock. Saturn is so light that it would float in water! It has 146 moons, including Titan which has its own atmosphere.

● Uranus

An ice giant that spins on its side — its axis is tilted almost 98 degrees! Uranus has 27 known moons, all named after characters from Shakespeare's plays.

● Neptune

The windiest planet. Winds on Neptune can reach 2,100 kilometres per hour — faster than any storm on Earth. Neptune is the most distant planet.

2.4 Dwarf Planets

Not everything that orbits the Sun is a planet. Some objects are too small. Astronomers call them dwarf planets. The most famous is Pluto.

Dwarf Planet

A dwarf planet orbits the Sun and is round, but it has not cleared its orbit of other objects. Pluto is the most famous dwarf planet.

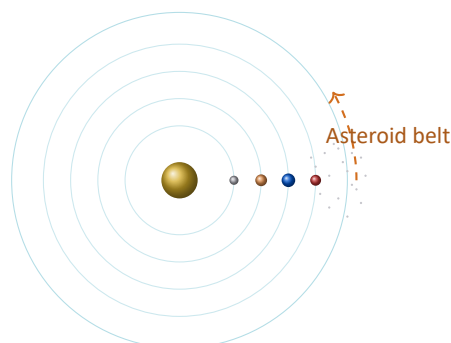
Poor Pluto!

Pluto was considered the ninth planet from 1930 until 2006, when astronomers decided it should be called a dwarf planet instead. It is smaller than Earth's Moon!

There are five recognised dwarf planets: **Pluto** (the largest), **Ceres** (in the asteroid belt), **Eris**, **Makemake**, and **Haumea**. Astronomers think there may be hundreds more!

2.5 Asteroids

Asteroids are rocky objects that orbit the Sun. They are much smaller than planets. Most are found in the **asteroid belt** between Mars and Jupiter.



The asteroid belt contains millions of rocky leftovers

Figure 2.2: The asteroid belt lies between Mars and Jupiter.

Leftover Building Blocks

Asteroids are leftovers from when the solar system formed, 4.6 billion years ago. They are pieces of a planet that never formed because Jupiter's gravity kept them from clumping together.

2.6 Comets — Dirty Snowballs

Comets are chunks of ice, dust, and rock that orbit the Sun in long, stretched-out paths. When a comet gets close to the Sun, the ice turns into gas and creates a beautiful glowing tail.

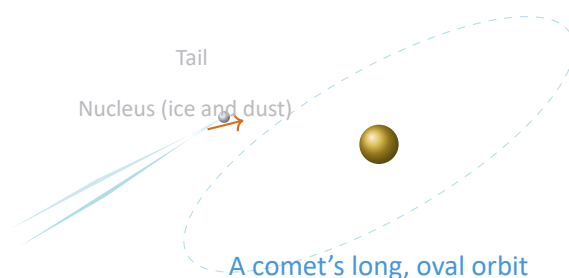


Figure 2.3: Comets have long, stretched-out orbits and grow tails near the Sun.

Halley's Comet

The most famous comet is Halley's Comet. It visits the inner solar system every 75 to 76 years. It was last seen in 1986 and will return in 2061. Mark your calendar!

Make a Model Solar System

Create a model of the solar system using items from home. Use a large ball or balloon for the Sun. Use smaller balls, beads, or play-dough for the planets. Arrange them in the

correct order. Label each planet with its name and one interesting fact. **Challenge:** Can you get the distances right? Remember that the outer planets are very far apart!

2.7 Chapter Review

Key Ideas

- The **solar system** has the Sun, eight planets, dwarf planets, asteroids, and comets.
- The four **inner planets** are small and rocky. The four **outer planets** are large and gaseous.
- **Dwarf planets** like Pluto are round worlds that orbit the Sun but are too small to be planets.
- **Asteroids** are rocky leftovers, mostly in the asteroid belt.
- **Comets** are icy bodies that grow tails when they approach the Sun.

Quick Questions

1. Name the eight planets in order from the Sun.
2. What is the difference between inner and outer planets?
3. Why was Pluto reclassified as a dwarf planet?
4. Where is the asteroid belt located?
5. Why do comets have tails?
6. Which planet is the biggest? Which is the smallest?

Words to Know

Asteroid A small rocky object orbiting the Sun.

Comet An icy object that grows a tail near the Sun.

Dwarf Planet A round object that orbits the Sun but is not big enough to be a planet.

Gas Giant A large planet made mostly of gas (Jupiter, Saturn).

Ice Giant A planet made mostly of ice and rock (Uranus, Neptune).

Year The time a planet takes to orbit the Sun once.

Explore More

- **NASA Solar System Exploration:** <https://solarsystem.nasa.gov>
- **ESA Kids: Our Solar System:** <https://www.esa.int/kids>
- **Make a comet in a cup:** <https://spaceplace.nasa.gov/comet>
- **Scale model of the solar system:** joshworth.com/pixelspace



PART II

Everything in the universe is in motion — including us!

EVERY day, the Sun rises and sets. Every night, the Moon changes shape and the stars slowly move across the sky. But nothing is actually moving around us — we are the ones moving! Earth is spinning and travelling through space.

Why does the sky look different every day and night?

3.1 Earth Spins (Rotation)

Imagine standing on a giant ball that slowly spins. That is exactly what Earth does! It rotates around an imaginary line called its **axis**.

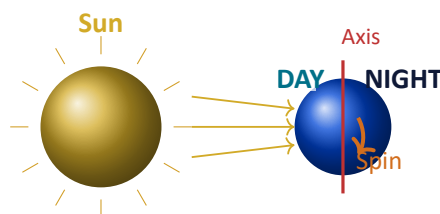


Figure 3.1: Earth spins on its axis. The side facing the Sun has day; the other side has night.

Rotation

Rotation is the spinning of an object around its own axis. One complete rotation of Earth takes 24 hours — that is one day.

Faster Than a Jet

Earth spins at about 1,670 kilometres per hour at the equator! But we don't feel it because everything around us — the air, the ground, the buildings — spins with us.

Why the Sun Rises and Sets

Earth spins from west to east. That is why the Sun appears to rise in the east and set in the west. When you watch the sunrise, you are watching your part of Earth turn toward the Sun. When you watch the sunset, you are watching your part of Earth turn away.

Different Time, Different Place

When it is daytime in one country, it is nighttime on the opposite side of the world. A child in Australia might be eating breakfast while a child in Brazil is sleeping!

3.2 Earth Travels (Revolution)

While Earth spins, it also travels around the Sun. This journey takes one full year.

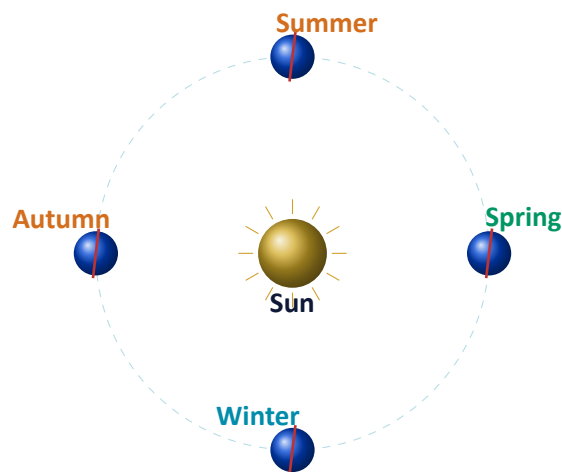
Revolution

Revolution is the movement of one object around another. Earth's revolution around the Sun takes about 365.25 days — one year.

Earth moves at an amazing 107,000 kilometres per hour — that is 30 kilometres every second! Every four years, we add an extra day (29 February) to make up for the extra quarter-day. This is called a **leap year**.

3.3 Why We Have Seasons

Why is summer warm and winter cold? Many people think it is because Earth is closer to the Sun in summer. **That is not true!** The seasons happen because Earth's axis is tilted.



Earth's tilted axis causes the seasons

Figure 3.2: Earth's axis is tilted 23.5° . This tilt causes summer, autumn, winter, and spring.

Earth's axis is tilted by 23.5 degrees. When the Northern Hemisphere tilts *toward* the Sun, it gets more direct sunlight — that is summer. When it tilts *away*, sunlight is spread out more — that is winter.

It Is Not the Distance!

Many people think summer is hot because Earth is closer to the Sun. Actually, Earth is closest to the Sun in January (winter for the Northern Hemisphere)! The tilt is what matters, not the distance.

Make Your Own Seasons

Use a torch (flashlight) and a ball in a dark room. Tilt the ball toward the torch — see how the light hits one part directly? That is summer! Tilt it away — now the light spreads out. That is winter! Try spinning the ball while keeping it tilted. What do you notice?

3.4 Why the Moon Changes Shape

Have you noticed that the Moon does not always look the same? Sometimes it is a thin crescent, sometimes half-lit, and sometimes a full bright circle. These are called **moon phases**.

The Moon orbits Earth about once a month. As it moves, we see different amounts of its sunlit side.

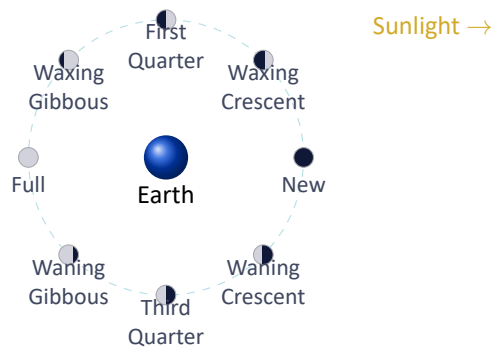


Figure 3.3: The eight phases of the Moon, as seen from above Earth.

New Moon The Moon is between Earth and the Sun. We cannot see it.

Waxing Crescent A thin sliver appears on the right side.

First Quarter Half of the Moon is lit — the right half.

Waxing Gibbous More than half lit, still growing.

Full Moon The whole face of the Moon is bright.

Waning Gibbous More than half lit, but shrinking.

Third Quarter The left half is lit.

Waning Crescent A thin sliver remains on the left.

Waxing vs Waning

A helpful trick: when the Moon is **waxing** (growing), the lit side is on the right. When it is **waning** (shrinking), the lit side is on the left. In the Northern Hemisphere, the right side gets light first!

3.5 Eclipses — When Shadows Fall

Sometimes the Sun, Earth, and Moon line up perfectly. When this happens, we get an eclipse — one of the most amazing sights in the sky!

Solar Eclipse

A solar eclipse happens when the Moon passes between Earth and the Sun. The Moon blocks the Sun's light, casting a shadow on Earth. For a few minutes, day turns into twilight!

Never Look at a Solar Eclipse!

Even during an eclipse, the Sun is still dangerously bright. Never look at a solar eclipse without special eclipse glasses. Regular sunglasses will NOT protect your eyes!

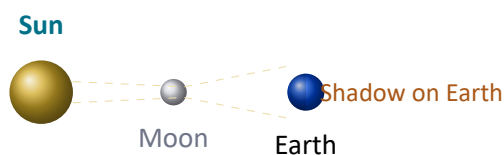


Figure 3.4: During a solar eclipse, the Moon blocks the Sun.

Lunar Eclipse

A lunar eclipse happens when Earth passes between the Sun and the Moon. Earth's shadow falls on the Moon, making it look dark and reddish.

Safe to Watch

Unlike solar eclipses, lunar eclipses are perfectly safe to watch with your eyes! They happen about twice a year and can be seen from anywhere on the night side of Earth.

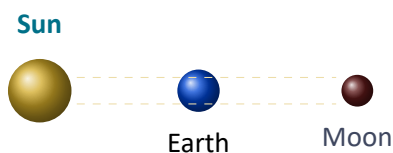


Figure 3.5: During a lunar eclipse, the Moon turns red.

During a total lunar eclipse, the Moon often turns a deep red colour. This is because some sunlight bends through Earth's atmosphere and reaches the Moon. The blue light scatters away, leaving only red light. That is why lunar eclipses are sometimes called **blood moons**.

Observe the Moon for a Month

For one month, look at the Moon every clear night. Draw its shape in a notebook with the date and time. At the end of the month, look at your drawings in order. Can you see the pattern? Which phase did you start with? Which phase did you end with?

3.6 Chapter Review**Key Ideas**

- **Rotation** is Earth spinning on its axis — one spin = one day.
- **Revolution** is Earth orbiting the Sun — one orbit = one year.
- **Seasons** are caused by Earth's 23.5° axis tilt, not by distance from the Sun.
- **Moon phases** happen because we see different amounts of the Moon's sunlit side.
- A **solar eclipse** occurs when the Moon blocks the Sun.
- A **lunar eclipse** occurs when Earth blocks sunlight from reaching the Moon.

Quick Questions

1. What is the difference between rotation and revolution?
2. What causes the seasons? Why is summer hotter than winter?
3. Draw the eight phases of the Moon in order starting from new moon.
4. Why does a solar eclipse happen? Why is it dangerous to look at?
5. Why does the Moon turn red during a lunar eclipse?
6. How long does it take Earth to spin around once?

Words to Know

Axis An imaginary line through the centre of Earth.

Eclipse When one celestial body blocks light from another.

Leap Year A year with 366 days, happening every four years.

Lunar Eclipse When Earth's shadow falls on the Moon.

Phase The shape of the lit part of the Moon.

Solar Eclipse When the Moon's shadow falls on Earth.

Waning When the lit portion of the Moon is getting smaller.

Waxing When the lit portion of the Moon is getting larger.

Explore More

- **NASA Space Place: Phases of the Moon:** <https://spaceplace.nasa.gov/moon-phases/>
- **Watch solar eclipse videos:** <https://www.nasa.gov/eclipse>
- **Track daylight where you live:** <https://www.timeanddate.com/sun/>
- **Find your shadow length:** <https://spaceplace.nasa.gov/shadows/>



W^{HY} does the Moon orbit Earth? Why do we stay on the ground instead of floating away? The answer is **gravity** — an invisible force that pulls everything toward everything else.

How does gravity work, and how do we travel in space?

4.1 What Is Gravity?

Gravity is the force that pulls objects together. The more mass an object has, the stronger its gravity. The Sun has the strongest gravity in the solar system, which is why all the planets orbit around it.

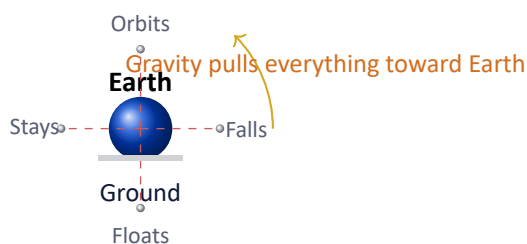


Figure 4.1: Gravity pulls everything toward the centre of Earth.

Gravity

Gravity is an invisible force that pulls objects with mass toward each other. The more mass an object has, the stronger its gravity. Earth's gravity keeps us on the ground and keeps the Moon in orbit.

Weight vs Mass

Your **mass** (how much matter is in your body) stays the same everywhere. Your **weight** (how strongly gravity pulls you) changes! On the Moon, you would weigh only one-sixth of your Earth weight. A 30 kg child would feel like just 5 kg on the Moon!

Gravity in the Solar System

The Sun's gravity holds the entire solar system together. Each planet's gravity keeps its moons in orbit. That is why everything stays in its place!

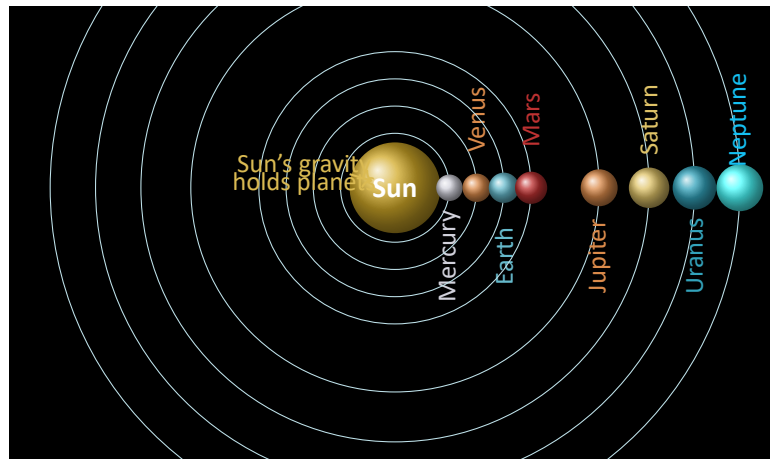


Figure 4.2: The Sun's gravity keeps the planets in orbit.

4.2 Orbits — Falling Around

An orbit happens when an object moves so fast sideways that it keeps missing the ground! Think of throwing a ball — it curves down. Now imagine throwing it so hard that the curve matches the curve of Earth. The ball would keep falling around Earth forever. That is an orbit!

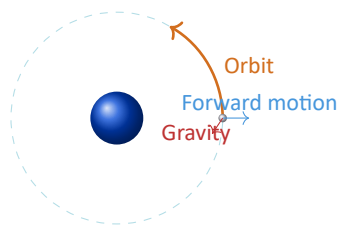


Figure 4.3: An orbit is a constant fall around Earth.

International Space Station

The ISS orbits Earth every 90 minutes! Astronauts on the ISS experience weightlessness not because there is no gravity, but because they are constantly falling around Earth. You can see the ISS pass overhead — it looks like a bright star moving across the sky!

4.3 How Rockets Work

Getting to space is hard because you have to fight gravity. That is why we use rockets — the most powerful machines ever built!



Figure 4.4: A rocket uses fuel to push itself upward.

Rockets work by **action and reaction** (Newton's Third Law). The engine burns fuel to create hot gases that blast out of the nozzle. The action is the gas shooting downward; the reaction is the rocket being pushed upward. This works even in space because the rocket carries its own oxygen!

Saturn V

The Saturn V rocket that launched astronauts to the Moon was as tall as a 36-storey building (111 metres)! Its five engines burned 15 tonnes of fuel per second.

Stages

Most rockets have **stages**: large fuel tanks that drop off when empty to make the rocket lighter. This is like a runner shedding a heavy backpack to go faster!

4.4 Exploring the Solar System

Humans have sent spacecraft to every planet in the solar system. Here are some of the most amazing missions:

Apollo 11 — 1969

The first mission to land humans on the Moon. Astronauts Neil Armstrong and Buzz Aldrin walked on the lunar surface while Michael Collins orbited above. "One small step for a man, one giant leap for mankind."

Voyager — 1977

Two spacecraft that explored Jupiter, Saturn, Uranus, and Neptune. Voyager 1 has now left the solar system and is travelling in interstellar space! Each Voyager carries a Golden Record with sounds and images of Earth.

Mars Rovers

Since 1997, NASA has sent rovers to explore Mars. Spirit and Opportunity arrived in 2004 — Opportunity lasted 14 years! Perseverance landed in 2021 carrying a small helicopter called Ingenuity, the first aircraft to fly on another planet.

Spacecraft Speed

The fastest spacecraft ever built, the Parker Solar Probe, travels at 690,000 kilometres per hour! That is fast enough to fly from London to New York in less than a minute.

4.5 Life in Space

What is it like to live in space? Astronauts on the ISS experience **microgravity** (almost weightless). Everything floats! They have to be strapped down to sleep, and they exercise two hours every day to keep their muscles strong.



Figure 4.5: The ISS — a home for astronauts in space.

Space Debris

There are thousands of pieces of old satellites and rocket parts orbiting Earth. This is called **space debris**. Even a tiny piece can damage a spacecraft because it moves so fast. Scientists are working on ways to clean up space junk!

Find the ISS Online

Go to <https://spotthestation.nasa.gov> and enter your location. Find out when the ISS will pass over your city tonight. Go outside at the right time and look for a bright, steady light moving across the sky. That is the ISS with astronauts inside!

4.6 Chapter Review

Key Ideas

- **Gravity** pulls everything together. More mass = stronger gravity.
- An **orbit** happens when an object moves sideways fast enough to keep missing the ground.
- **Rockets** work by action and reaction: hot gas shoots down, the rocket goes up.
- **Spacecraft** have visited every planet. The Voyager probes have left the solar system.
- Astronauts on the **ISS** live in microgravity, floating inside the station.

Quick Questions

1. What is gravity, and what does it depend on?
2. Why do astronauts float on the ISS?
3. How does a rocket work? What is the action and reaction?
4. Which space mission first landed humans on the Moon?
5. What is space debris, and why is it a problem?
6. Would you weigh more or less on the Moon? Why?

Words to Know

Action and Reaction Newton's Third Law: every action has an equal opposite reaction.

Gravity The invisible force that pulls objects with mass together.

ISS International Space Station — a laboratory orbiting Earth.

Microgravity A condition where objects appear to be weightless.

Orbit The curved path of an object around another in space.

Rocket A vehicle that uses thrust to travel into space.

Space Debris Old satellites and rocket parts orbiting Earth.

Explore More

- **NASA Space Place: Gravity:** <https://spaceplace.nasa.gov/gravity/>
- **Spot the ISS:** <https://spotthestation.nasa.gov>
- **ESA Kids: Space for Kids:** <https://www.esa.int/kids>
- **NASA Rocket Simulator:** <https://www.grc.nasa.gov/www/k-12/rocket/>
- **Watch rocket launches live:** <https://www.nasa.gov/nasatv>



PART III

Look up at the night sky and journey to the farthest reaches of the universe.

ON a clear, dark night, look up. Hundreds of tiny lights sparkle in the sky. Each one is a sun — a giant ball of hot gas, just like our own Sun, but incredibly far away. Let us get to know the stars.

What are stars, and how can we find our way around the night sky?

5.1 What Is a Star?

A star is a huge ball of hot gas that makes its own light. Deep inside a star, hydrogen atoms smash together to make helium. This is called **nuclear fusion**, and it releases enormous amounts of energy. That energy travels through space as light and heat.

Star

A star is a giant ball of hot gas that produces its own light and heat through nuclear fusion. Our Sun is a star.

Star Colours

Stars come in different colours! The colour tells you how hot the star is.



Figure 5.1: Star colours from hottest (blue) to coolest (red).

Our Sun is a **yellow** star, right in the middle. Blue stars are extremely hot (over 30,000°), and red stars are cooler (about 3,000°).

The Nearest Star

The closest star to Earth (after the Sun) is Proxima Centauri, 4.2 light-years away. Light from Proxima Centauri takes 4.2 years to reach us. If you travelled on the fastest rocket ever built, it would take about 70,000 years to get there!

Star Sizes

Stars come in different sizes too. Some are tiny red dwarfs, only a bit bigger than Jupiter. Some are enormous supergiants — if one replaced our Sun, it would reach past the orbit of Mars!

How Many Stars?

On a clear, dark night, you can see about 2,500 stars with your eyes alone. But in the universe, there are more stars than grains of sand on every beach on Earth!

5.2 Constellations — Pictures in the Sky

Long ago, people connected the dots between stars to make pictures. These are called **constellations**. Different cultures saw different pictures — while the ancient Greeks saw a hunter (Orion), the Chinese saw a different pattern in the same stars.

There are 88 official constellations that help astronomers navigate the sky.

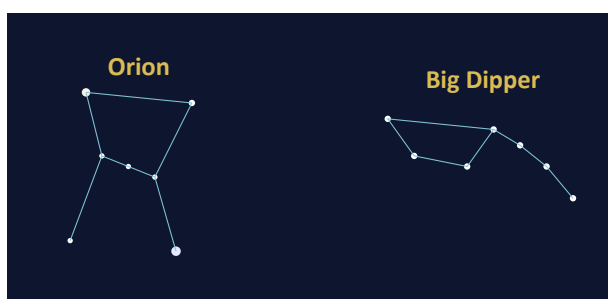


Figure 5.2: Two famous constellations: Orion (left) and the Big Dipper (right).

Finding North with the Big Dipper

The Big Dipper can help you find the North Star (Polaris). The two stars at the end of the dipper's bowl point directly to Polaris. Polaris stays almost still in the sky while all the other stars appear to rotate around it.

88 Constellations

There are 88 official constellations. Some famous ones include: **Ursa Major** (Big Dipper), **Orion** (the Hunter), **Cassiopeia** (the Queen), **Scorpius** (the Scorpion), and **Cygnus** (the Swan).

The Zodiac

You may have heard of the zodiac signs — Aries, Taurus, Gemini, and so on. These are actually constellations that lie along the path the Sun takes across the sky (called the **ecliptic**). When someone says they are a **Leo**, it means the Sun was in the constellation Leo on the day they were born!

5.3 Galaxies — Star Cities

Stars are not scattered randomly through space. They are grouped into huge collections called **galaxies**. A galaxy can contain billions or even trillions of stars.

Galaxy

A galaxy is a huge group of stars, gas, dust, and dark matter held together by gravity. Our galaxy is the Milky Way.

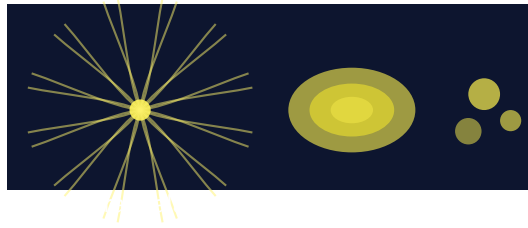


Figure 5.3: Three types of galaxies: spiral, elliptical, and irregular.

Our galaxy, the **Milky Way**, is a spiral galaxy. On a very dark night, you can see it as a faint, milky band of light stretching across the sky. The Milky Way contains over 100 billion stars. And there are billions of other galaxies in the universe!

5.4 Our Galactic Home

The Milky Way is a huge spinning disc of stars. Our Sun is located about halfway from the centre, on one of the spiral arms. It takes the Sun about 230 million years to complete one orbit around the centre of the galaxy!

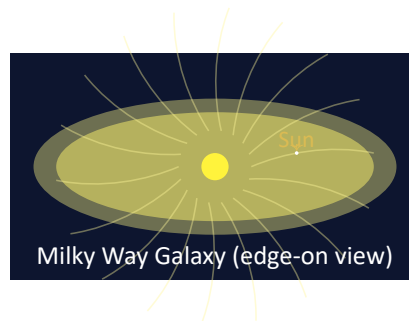


Figure 5.4: The Milky Way seen from the side — a flat disc with a bright centre.

5.5 Tips for Stargazing

You do not need a telescope to enjoy astronomy! Here are some things you can do with just your eyes:

Find Orion at Night

On a clear winter evening, look south (Northern Hemisphere). Find three bright stars in a straight line — that is Orion's Belt. Above the belt are two bright stars (his shoulders). Below are two more (his knees). Can you see the whole hunter?

- **Find the Big Dipper** — Look north. It looks like a giant ladle.

- **Count the Pleiades** — A small cluster of stars that looks like a tiny dipper.
- **Watch for shooting stars** — Meteors streak across the sky, especially during meteor showers.
- **See the Milky Way** — On a very dark night, far from city lights, look for a faint band of light across the sky.

Light Pollution

City lights make it hard to see stars. This is called **light pollution**. If you can, visit a dark place away from city lights — the difference is amazing! In a truly dark sky, you can see thousands of stars.

Create a Star Wheel

Make a simple planisphere (star finder) to identify constellations any night of the year. Print one from <https://spaceplace.nasa.gov/starfinder/> and follow the instructions. Take it outside and see how many constellations you can name!

5.6 Chapter Review

Key Ideas

- **Stars** are giant balls of hot gas that make their own light. Our Sun is a star.
- Star **colour** tells you the temperature: blue is hottest, red is coolest.
- **Constellations** are patterns of stars that help us navigate the sky.
- **Galaxies** are huge collections of stars. Our galaxy is the Milky Way.
- With just your eyes, you can see stars, planets, constellations, and the Milky Way.

Quick Questions

1. What makes a star shine?
2. Which type of star is hotter: blue or red?
3. What is a constellation? How many official constellations are there?
4. What is the nearest star to Earth (after the Sun)?
5. Name three types of galaxies.
6. Why is it harder to see stars in a city than in the countryside?

Words to Know

Constellation A group of stars that forms a pattern in the sky.

Galaxy A huge group of stars, gas, and dust held together by gravity.

Light Pollution Bright city lights that make it hard to see stars.

Milky Way The spiral galaxy we live in.

Nuclear Fusion The process that powers stars, turning hydrogen into helium.

Planisphere A rotating star chart used to identify constellations.

Red Dwarf A small, cool, dim star.

Supergiant An extremely large, bright star.

Explore More

- NASA Space Place: Stars: <https://spaceplace.nasa.gov/stars/>
- Find constellations tonight: <https://www.skyatnightmagazine.com>
- Stellarium Web (free online planetarium): <https://stellarium-web.org>
- Hubble images of stars: <https://hubblesite.org>
- Light pollution map: <https://www.lightpollutionmap.info>



W^E live on a planet that orbits a star. That star is one of billions in the Milky Way galaxy. And the Milky Way is just one galaxy among billions in the universe. Let us explore what lies beyond our own galaxy.

What is beyond our solar system, and how did the universe begin?

6.1 Our Galaxy: The Milky Way

The Milky Way is a spiral galaxy — a huge, spinning disc of stars, gas, and dust. It is so big that light takes 100,000 years to travel from one side to the other!

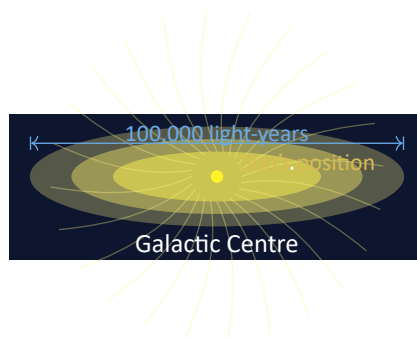


Figure 6.1: The Milky Way is a spiral galaxy. Our Sun is about halfway out from the centre.

Light-Year

A light-year is the distance light travels in one year — about 9.5 trillion kilometres. When we say a star is 10 light-years away, we mean its light takes 10 years to reach us.

Galactic Year

The Sun takes about 230 million years to complete one orbit around the centre of the Milky Way. The last time the Sun was in its current position, dinosaurs had just appeared on Earth!

What Is at the Centre?

At the centre of the Milky Way is a **supermassive black hole** called Sagittarius A* (pronounced “Sagittarius A-star”). It has a mass of about 4 million Suns! Despite its name, a black hole does not suck in everything around it — stars can orbit safely at a distance.

6.2 Other Galaxies

The Milky Way is not alone. It is part of a small group of galaxies called the **Local Group**, which includes the Andromeda Galaxy (our largest neighbour) and about 50 smaller galaxies.

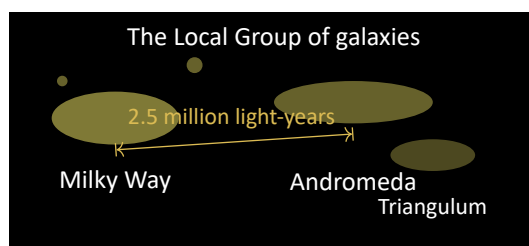


Figure 6.2: The Local Group contains the Milky Way, Andromeda, and dozens of smaller galaxies.

Galaxies come in different shapes:

- **Spiral galaxies** — like the Milky Way, with beautiful swirling arms.
- **Elliptical galaxies** — smooth and oval-shaped, with little gas or dust.
- **Irregular galaxies** — no clear shape, often from collisions.

Galaxies Collide!

The Milky Way and Andromeda are moving toward each other. In about 4.5 billion years, they will crash and merge into one giant galaxy! Do not worry — stars are so far apart that most will not collide. Our Sun might be flung to a new part of the merged galaxy.

6.3 The Expanding Universe

In the 1920s, astronomer Edwin Hubble made an amazing discovery: galaxies are moving away from us! The farther away a galaxy is, the faster it is moving. This means the universe is **expanding**.

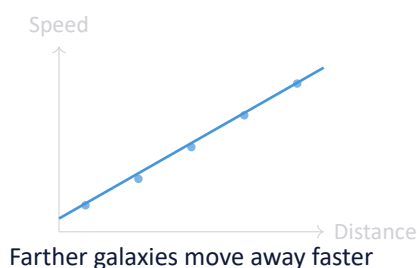


Figure 6.3: Hubble’s discovery: galaxies farther from us move away faster.

Imagine a loaf of raisin bread rising in the oven. As the bread expands, every raisin moves away from every other raisin. The universe is like that — space itself is stretching, carrying galaxies along with it.

Hubble Space Telescope

The Hubble Space Telescope, named after Edwin Hubble, has been orbiting Earth since 1990. It has taken the most detailed images of distant galaxies ever seen, including the famous Hubble Ultra Deep Field showing thousands of galaxies in a tiny patch of sky.

6.4 The Beginning: The Big Bang

If the universe is expanding, then in the past everything must have been closer together. This leads to a remarkable idea: the universe began in a single, tiny point about 13.8 billion years ago. This is called the **Big Bang**.

It Was Not an Explosion!

The Big Bang was not an explosion in space — it was the beginning of space and time themselves. There was no “before” the Big Bang because time began with it.

Evidence for the Big Bang

1. **The universe is expanding** (Hubble’s discovery).
2. **Cosmic Microwave Background** — faint radio waves from the early universe, like an echo of the Big Bang.
3. **The right mix of elements** — the universe has exactly the amount of hydrogen and helium that the Big Bang predicts.



Figure 6.4: From the Big Bang to today — the universe keeps expanding.

CMB: Echo of the Big Bang

The Cosmic Microwave Background (CMB) is faint radiation that fills the entire universe. It was discovered in 1965 and is the oldest light we can see — coming from just 380,000 years after the Big Bang. It is like a baby picture of the universe!

6.5 The Mystery Ingredients

Everything we can see — stars, planets, galaxies, people — makes up only about 5% of the universe. The rest is invisible and mysterious.

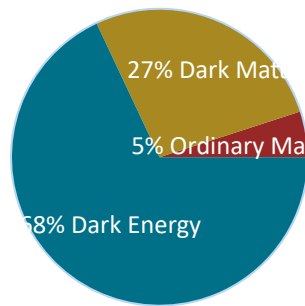


Figure 6.5: What the universe is made of. Most of it is invisible!

Dark Matter An invisible substance that does not emit light but has gravity. We know it exists because galaxies spin too fast to be held together by visible matter alone.

Dark Energy A mysterious force that is causing the universe to expand faster and faster. Scientists are still trying to understand what it is!

A Universe of Mysteries

Scientists estimate that 95% of the universe is invisible to us — dark matter and dark energy. What if there are whole other types of matter out there that we cannot detect with our current instruments? How would you design an experiment to find something you cannot see?

Make Your Own Expanding Universe

Take a balloon. Draw dots on it with a marker (each dot is a galaxy). Now blow up the balloon. Watch what happens to the dots — they all move away from each other! That is how the expanding universe works. The farther two dots are, the faster they separate.

6.6 Chapter Review

Key Ideas

- The **Milky Way** is a spiral galaxy containing over 100 billion stars.
- A **light-year** is the distance light travels in one year.
- The **universe is expanding** — galaxies are moving away from each other.
- The **Big Bang** started the universe 13.8 billion years ago.
- Most of the universe is made of invisible **dark matter** and **dark energy**.

Quick Questions

1. What type of galaxy is the Milky Way?
2. How long does it take light to cross the Milky Way?

3. What did Edwin Hubble discover about galaxies?
4. What is the Big Bang, and how long ago did it happen?
5. What is the Cosmic Microwave Background?
6. How much of the universe is made of ordinary matter (the stuff we can see)?

Words to Know

Big Bang The event that began the universe 13.8 billion years ago.

Black Hole A region of space with gravity so strong that nothing can escape.

Cosmic Microwave Background Faint radiation left over from the Big Bang.

Dark Energy A mysterious force causing the universe to expand faster.

Dark Matter Invisible matter that has gravity but does not emit light.

Light-Year The distance light travels in one year.

Local Group The small group of galaxies containing the Milky Way.

Supernova A massive explosion at the end of a star's life.

Explore More

- NASA Space Place: Universe: <https://spaceplace.nasa.gov/universe/>
- Hubble Space Telescope images: <https://hubblesite.org>
- James Webb Space Telescope: <https://webbtelescope.org>
- Big Bang timeline: <https://www.esa.int/kids>
- What is dark matter? <https://home.cern/science/physics/dark-matter>



Planet	Distance (million km)	Diameter (km)	Day Length	Year Length	Moons
Mercury	57.9	4,879	59 Earth days	88 days	0
Venus	108.2	12,104	243 Earth days	225 days	0
Earth	149.6	12,742	24 hours	365.25 days	1
Mars	227.9	6,779	24.6 hours	687 days	2
Jupiter	778.5	139,820	9.9 hours	11.9 years	95
Saturn	1,434	116,460	10.7 hours	29.5 years	146
Uranus	2,871	50,724	17.2 hours	84 years	28
Neptune	4,495	49,244	16.1 hours	165 years	16

- **Inner planets:** Mercury, Venus, Earth, Mars — small, rocky, close to the Sun.
- **Outer planets:** Jupiter, Saturn, Uranus, Neptune — large, gaseous, far from the Sun.

Name	Uppercase	Lowercase
Alpha	A	α
Beta	B	β
Gamma	Γ	γ
Delta	Δ	δ
Epsilon	E	ϵ
Zeta	Z	ζ
Eta	H	η
Theta	Θ	θ
Iota	I	ι
Kappa	K	κ
Lambda	Λ	λ
Mu	M	μ
Nu	N	ν
Xi	Ξ	ξ
Omicron	O	o
Pi	Π	π
Rho	P	ρ
Sigma	Σ	σ
Tau	T	τ
Upsilon	Υ	v
Phi	Φ	ϕ
Chi	X	χ
Psi	Ψ	ψ
Omega	Ω	ω

Quantity	Symbol	Value
Speed of Light	c	$3.00 \times 10^8 \text{ m/s}$
Gravitational Constant	G	$6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
Astronomical Unit	AU	$1.50 \times 10^{11} \text{ m}$ (Sun–Earth distance)
Light-Year	ly	$9.46 \times 10^{15} \text{ m}$
Solar Mass	$M_{\odot}$	$1.99 \times 10^{30} \text{ kg}$
Solar Radius	$R_{\odot}$	$6.96 \times 10^8 \text{ m}$
Earth Mass	$M_{\oplus}$	$5.97 \times 10^{24} \text{ kg}$
Earth Radius	$R_{\oplus}$	$6.37 \times 10^6 \text{ m}$
Hubble Constant	H_0	$\approx 70 \text{ km/s/Mpc}$

Prefix	Symbol	Factor
Giga	G	10^9 (billion)
Mega	M	10^6 (million)
Kilo	k	10^3 (thousand)
Centi	c	10^{-2} (hundredth)
Milli	m	10^{-3} (thousandth)
Micro	μ	10^{-6} (millionth)
Nano	n	10^{-9} (billionth)

Newton's Law of Gravity $F = G \frac{Mm}{r^2}$

Gravitational force between two objects

Speed = Distance / Time $v = \frac{d}{t}$

How fast something is moving

Density = Mass / Volume $\rho = \frac{M}{V}$

How compact something is

Distance from Speed of Light $d = c \times t$

Light travel distance

Property	Sun	Moon
Mass	$1.99 \times 10^{30} \text{ kg}$	$7.34 \times 10^{22} \text{ kg}$
Radius	$6.96 \times 10^8 \text{ m}$	$1.74 \times 10^6 \text{ m}$
Surface Temperature	$5,500^\circ$	$-180 \text{ to } 120^\circ$
Distance from Earth	150 million km	384,400 km
Age	4.6 billion years	4.5 billion years

Data from NASA/JPL. Current as of 2026.